

MTH202 ENDSEM EXAM

Max marks: 80

Total time: 3 hrs.

Notes:

- (1) All variables a, b, c, y, z, r, i, j should be taken to have values in integers, unless specified otherwise.
- (2) The symbol x denotes the variable, unless otherwise noted.
- (3) You can use any result we proved in class provided you clearly mention the same.

Problem 1 ($2 \times 6 = 12$). Answer the following with one line explanation.

- (1) What remainder does $15!$ leave when divided by 17 ?
- (2) If f and g are totally multiplicative arithmetic functions, then their Dirichlet convolution $f * g$ is also totally multiplicative.
- (3) What is the order of integers 2 modulo 11?
- (4) State the condition (without proof) on integers a, b, d such that linear diophantine equation $ax + by = d$ has at least one solution.
- (5) Assume p is a prime and $p \geq 3$. Assume $p \nmid a$. Is the order of a^{p-2} always the same as the order of $a \pmod{p}$? (True/False).
- (6) Is 3 a perfect square $\pmod{23}$? (True/False)

Problem 2 (3×4). Answer the following with a short argument:

- (1) Let p be an odd prime. Show that if $p \mid ab$ then $p \mid a$, or $p \mid b$.
- (2) Let p be an odd prime. Show that for any integer x , we have:

$$\prod_{0 \leq i < j \leq p-1} (x^i - x^j) \equiv 0 \pmod{p}.$$

- (3) If n has a primitive root, how many incongruent $\pmod{n}$ primitive roots does it have? Express your answers in terms of the Euler's function ϕ .
- (4) Compute multiplicative inverse of 3 $\pmod{97}$, that is an integer n such that $3n \equiv 1 \pmod{97}$. Also compute $3^{95} \pmod{97}$. Why should they be the same?

Problem 3 (6 points). Find *all* the solutions of following system of linear congruences using the Chinese Remainder Theorem. If no solution exists, state why.

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}.$$

Problem 4 ($2 + 4 = 6$ points). Suppose that you are given that $7^6 + 3^6 + 19$ is not a prime. Conclude that $p^6 + q^6 + 19$ is never a prime if p, q are primes.

Problem 5 ($3 + 4 = 7$ points). Let n be any positive integer. Show that:

- (1) The integers $a = n^3 + 2n$ and $b = n^2 + 1$ have greatest common divisor as 1.
- (2) Find some integers u and v (which may depend on n or be constants) such that:

$$(n^3 + 2n)u + (n^2 + 1)v = 1.$$

Problem 6 ($2 + 3 + 3 = 8$ points). Compute the following:

- (1) Compute $\phi(360)$, where ϕ is the Euler's totient function.
- (2) Use Euler's Theorem to compute the last two digits of 7^{362} .
- (3) Find $\left(\frac{15}{71}\right)$ (here 71 is a prime number but 15 is not!).

Problem 7 ($3 + 1 + 5 = 9$ points). Let $\tau(n)$ denote the number of positive divisors of n .

- (1) Show that $\tau(n)$ is an odd integer if and only if n is a perfect square.
- (2) Find smallest integer n with $\tau(n) = 10$.
- (3) Let $\sigma(n)$ to be the sum of divisors of n , and $\phi(n)$ be Euler's totient function, as before. Show that:

$$\sigma(n) = \sum_{d|n} \phi(d) \tau\left(\frac{n}{d}\right).$$

Problem 8 ($3 + 2 + 5 = 10$ points). Let $p > 3$ be an odd prime. Assume that there is an integer a such that $a^2 + a + 1 \equiv 0 \pmod{p}$. Show that:

- (1) Show that $a^4 + a^2 + 1 \equiv 0 \pmod{p}$ and hence $a \not\equiv a^2 \pmod{p}$ are the only two integers (upto $\equiv \pmod{p}$) satisfying the equation $x^2 + x + 1 = 0 \pmod{p}$.
- (2) Show that $p \equiv 1 \pmod{3}$.
- (3) Show that number of integers r in $\{1, 2, \dots, p-1\}$ such that $x^3 \equiv r \pmod{p}$ has a solution for x is $\frac{p-1}{3}$.

Problem 9 ($2 + 5 + 3 = 10$ points). Let p, q be odd primes such that $p = 2q + 1$. Prove that

- (1) if $\left(\frac{a}{p}\right) = 1$ then a cannot be a primitive root $\pmod{p}$.
- (2) if $\left(\frac{a}{p}\right) = -1$ and $a \not\equiv -1 \pmod{p}$ then a is a primitive root $\pmod{p}$.
- (3) Now use quadratic reciprocity and parts above to find out if 7 is a primitive root $\pmod{11}$.